Description

Image





Caption

1. Hair dryer with a Nichrome heating element. an alloy of nickel and chromium. © ANSYS, Inc. 2. Toaster with a Nichrome heating element. an alloy of nickel and chromium. © ANSYS, Inc.

The material

Nickel forms a wide range of alloys, valued by the chemical engineering and food processing industries for their resistance to corrosion, and by the makers of furnaces and high temperature equipment for their ability to retain useful strength at temperatures up to 1200 C. Typical of these are the nickel-chromium (Ni-Cr) alloys, often containing some iron (Fe) as well. The chromium increases the already-good resistance to corrosion and oxidation by creating a surface film of Cr2O3, the same film that makes stainless steel stainless. The data given here are for nickel-chromium alloys. There are separate records for stainless steel and nickel-based super alloys.

Compositional summary

Ni + 10 to 30% Cr + 0 to 10% Fe

General properties

Density		8.3e3	-	8.5e3	kg/m^3
Price	*	18.8	-	23.4	GBP/kg
Date first used		1905			

Mechanical properties

Young's modulus	200	-	220	GPa
Shear modulus	72	-	82	GPa
Bulk modulus	155	-	285	GPa
Poisson's ratio	0.305	-	0.315	
Yield strength (elastic limit)	365	-	460	MPa
Tensile strength	615	-	760	MPa

Compressive strength		365	-	460	MPa
Elongation		20	-	35	% strain
Hardness - Vickers		160	-	200	HV
Fatigue strength at 10^7 cycles	*	245	-	380	MPa
Fracture toughness	*	80	-	110	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	4e-4	-	0.002	

Thermal properties

Melting point		1.35e3	-	1.43e3	°C
Maximum service temperature	*	900	-	1e3	°C
Minimum service temperature		-272	-	-271	°C
Thermal conductor or insulator?		Poor conductor			
Thermal conductivity		9	-	15	W/m.°C
Specific heat capacity		430	-	450	J/kg.°C
Thermal expansion coefficient		12	-	14	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Poor conductor			
Electrical resistivity		102	-	114	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		3			
Formability		3	-	4	
Machinability		3			
Weldability		4	-	5	
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Values marked * are estimates.
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Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Acceptable
Nitric acid (10%)	Acceptable
Nitric acid (70%)	Limited use
Phosphoric acid (10%)	Acceptable
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Acceptable

Durability: alkalis

Sodium hydroxide (10%)	Acceptable
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Excellent
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Excellent
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.37e6	tonne/yr
Reserves, principal component	1.05e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 14.3	-	15.8	kg/kg
Embodied energy (virgin grade production)	* 228	-	252	MJ/kg
Water usage	* 245	-	270	l/kg

Material processing: energy

Metal casting, embodied energy	* 46.9	-	53.6	MJ/kg
Metal extrusion, hot, embodied energy	* 27.4	-	36.9	MJ/kg

Metal rolling and forging, embodied energy	* 14.3	-	19.1	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 56.9	-	71	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.06e4	-	1.23e4	MJ/kg
Machining, coarse, embodied energy	* 15.9	-	18.8	MJ/kg
Machining, fine, embodied energy	* 18.6	-	24.1	MJ/kg
Machining, grinding, embodied energy	* 21.3	-	29.4	MJ/kg
Machining, laser, CO2, embodied energy	* 1.26	-	1.46	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.37	-	3.87	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 1.79	-	2.42	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.936	-	1.25	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 4.63	-	5.78	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 869	-	1.01e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.04	-	1.23	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.22	-	1.58	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.39	-	1.93	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.103	-	0.12	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.86	-	2.05	kg/kg
Embodied energy, recycling	* 16	-	17.6	MJ/kg
Recycle fraction in current supply	48			%
Combust for energy recovery	✗			
Landfill	✗			
Biodegrade	✗			
Toxicity rating	Slightly toxic			
A renewable resource?	✗			

Environmental notes

Nickel alloys are non-toxic and can be recycled.

Supporting information

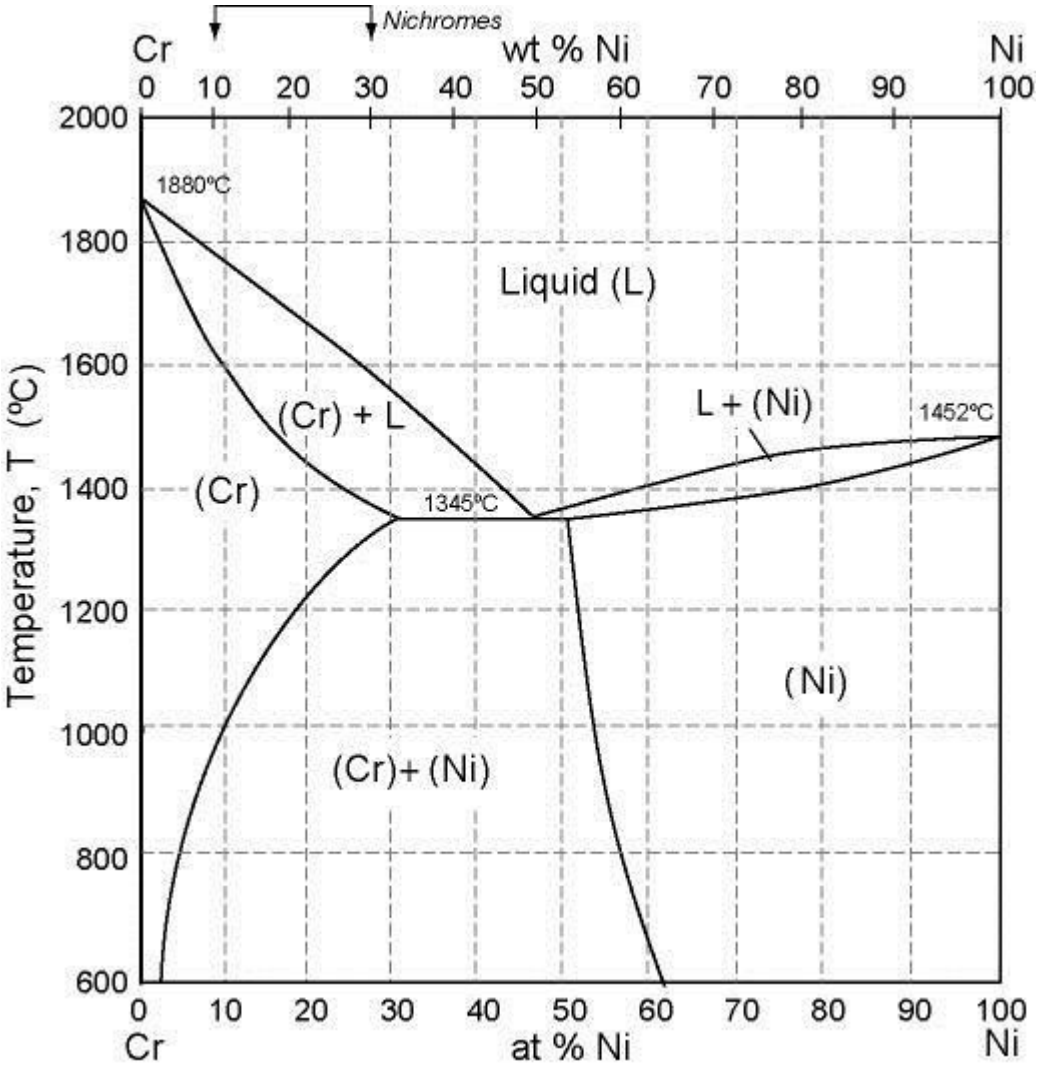
Design guidelines

Few other alloy systems can equal the nickel-based alloys in their combination of corrosion resistance and strength at low and high temperature. They are used in marine applications for heat exchanges in other structures. Nickel-iron alloys have high magnetic permeability (where electronic shielding and magnetic coils) and low thermal expansion (good for glass-to-metal joints). Invar, an alloy based on nickel, has essentially zero thermal expansion coefficient near room temperature; a magnetic contraction counteracts the ordinary thermal expansion, canceling it out. Nickel-chrome-iron alloys have high electrical resistance and are used as heating elements in toasters and industrial furnaces. Bi-metallic sheet of nickel bonded to copper is used as actuators for thermostats and safety devices. Alloys with titanium (Nitanols) have the remarkable property that they spring back to shape after severe deformation - they are called "shape memory alloys" for this reason.

Technical notes

Ni-Cr-Fe alloys are marketed under a bewildering catalog of tradenames. Chromels are straight Ni-Cr alloys with up to 20% Cr. Durimet and Nichromes contain iron. There are many more.

Phase diagram



Phase diagram description

Nichromes are alloys of nickel (Ni) with 10 - 30% chromium (Cr). They have useful strength and oxidation resistance to 1200 C.

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Typical uses

Heating elements and furnace windings, bi-metallic strips, thermocouples, springs, food processing equipment, chemical engineering equipment.

Links

ProcessUniverse

Producers

Reference

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